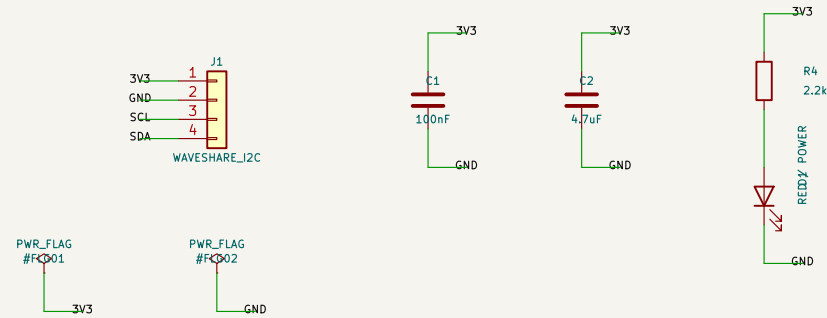


CoffeeFlow – screen sensor board / electrical review draft

3.3 V ONLY. Waveshare H8 must select 3V3. All matching labels are electrically connected.

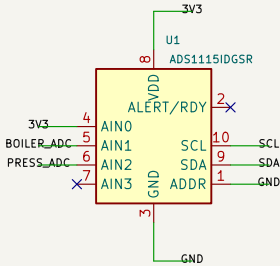
1 / Waveshare power and I2C



POWER: 3V3. No current does not verify I2C communication.

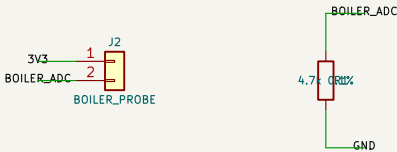
No added I2C pins: Waveshare H8 already has 4 pins.

2 / ADS1115 – 0x48



A0 = 3V3; A1 = boiler; A2 = pressure; A3 unused.

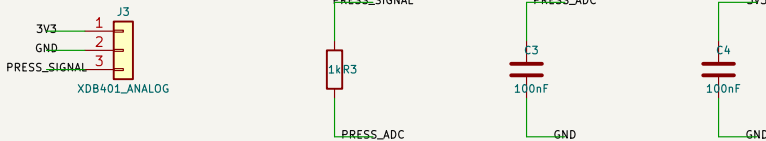
3 / Boiler NTC or PT1000



Probe between 3V3 and DIV. Fixed resistor from DIV to GND.

Rprobe = R1 * (A0/A1 – 1). Use new–board calibration.

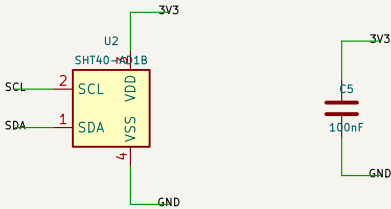
4 / XDB401 analog: 0.4–2.4 V / 0–12 bar



No voltage divider / amplifier needed. Pressure = 6*(Vout–0.4) bar nominal.

Verify supply current, output impedance, startup and ratiometry before manufacture.

5 / Enclosure humidity + temperature – 0x44



Place near board edge, exposed to box air, away from display heat.

Humidity sensor opening must remain free of flux, coating and adhesives.

Sheet: /
File: screen_sensors_draft.kicad_sch

Title:

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.6

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